

■ Custom Technology & Hardware Image Classifier

Comprehensive Academic & Technical Project Report

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Framework: TensorFlow / Keras 3 & Streamlit

Architecture: MobileNetV2 (Transfer Learning)

Repository: github.com/akshad-n/custom-Image-classifier

Live Demo: custom-image-classifier.streamlit.app

1. Executive Summary

This project presents an end-to-end Computer Vision system capable of accurately identifying and classifying 10 distinct technological objects and peripheral hardware components. Powered by a fine-tuned **MobileNetV2** deep neural network, the solution achieves high classification accuracy while maintaining a lightweight footprint (~20.4 MB model parameter payload), making it well-suited for both local execution and low-resource cloud hosting.

The model is deployed via a rich **Streamlit** web application that delivers instant image previews, top predicted labels, confidence metrics, and a Top-5 probability distribution breakdown. By employing in-memory caching (@st.cache_resource), inference latency is kept under 100 milliseconds per sample without repeated disk reloads.

2. Problem Statement & Objectives

Problem Statement: In hardware refurbishment facilities, automated inventory tracking, and e-waste processing, manual sorting of peripheral components is repetitive, labor-intensive, and error-prone. Variations in background, lighting, and orientation create major challenges for traditional rule-based inspection.

Core Objectives:

- **High-Accuracy Classification:** Accurately distinguish 10 common hardware categories under varied angles and lighting.
- **Lightweight Architecture:** Utilize depthwise separable convolutions to enable rapid CPU inference without dedicated GPU hardware.
- **Intuitive User Interface:** Offer an interactive UI with immediate visual feedback, confidence scoring, and model transparency.
- **Production Optimization:** Decouple class labels via JSON schemas and optimize container build footprints using tensorflow-cpu.

3. Dataset & Class Taxonomy

The classification taxonomy covers 10 hardware and peripheral classes. During preprocessing, images are standardized to **224 × 224 pixels** in 3-channel RGB. Class indices follow Keras's standard alphabetical order:

Index	Class Label	Description & Distinctive Visual Features
0	USB stick	Portable flash drive with rectangular body and metallic Type-A/C connector.
1	computer mouse	Ergonomic hand-held peripheral, curved housing, optical sensor, scroll wheel.
2	keyboard I	Input board featuring mechanical or membrane keycap grid and rectangular casing.
3	keys objects	Metal keys, ring attachments, serrated blade teeth, residential/office padlocks.
4	laptop	Portable clamshell computer, hinge, keyboard chassis, LCD/OLED panel.
5	magnifying glass	Circular biconvex optical lens with metal/plastic rim and elongated handle.
6	phone	Smartphone with touchscreen glass slab, narrow bezels, rear camera module.
7	router	Network gateway, external dipole antennas, LED indicators, RJ-45 ports.
8	satellite dish device	Parabolic reflector dish, feed horn, mounting bracket for telecommunications.
9	server rack	19-inch metal cabinet containing modular server blades and cable wiring.

4. Deep Learning Architecture & Pipeline

The model architecture is built upon **MobileNetV2** using transfer learning. MobileNetV2 uses **inverted residual blocks** and **depthwise separable convolutions**, substantially reducing computation and memory requirements while retaining rich semantic representations trained on ImageNet.

Layer (Type)	Output Shape	Parameters	Operational Role
input_layer_1	(None, 224, 224, 3)	0	Standardized 24-bit RGB image input.
data_augmentation	(None, 224, 224, 3)	0	Horizontal random flips for regularized invariance.
rescaling	(None, 224, 224, 3)	0	Normalizes [0, 255] pixel integers to [-1, 1].
mobilenetv2_1.00_224	(None, 7, 7, 1280)	2,257,984	Pre-trained convolutional feature extractor.
global_avg_pooling2d	(None, 1280)	0	Spatial 7×7 feature reduction into 1D embedding.
dropout	(None, 1280)	0	Random feature deactivation for anti-overfitting.
dense (Softmax)	(None, 10)	12,810	Class probability distribution over 10 categories.

Total Parameters: 5,349,216 (~20.41 MB)	Trainable Parameters: 1,539,210 (~5.87 MB)
Non-Trainable Params: 731,584 (~2.79 MB)	File Format: Serialized .keras archive

5. Experimental Validation & Results

The model was verified against ground-truth validation images extracted across each target class. The evaluation confirmed sharp probability separation with high confidence across all classes:

Test Target Class	Predicted Class	Argmax	Confidence	Validation Status
USB stick	USB stick	0	99.9%	✓ Verified
computer mouse	computer mouse	1	99.1%	✓ Verified
keyboard I	keyboard I	2	100.0%	✓ Verified
keys objects	keys objects	3	98.1%	✓ Verified
laptop	laptop	4	99.2%	✓ Verified
magnifying glass	magnifying glass	5	99.5%	✓ Verified
phone	phone	6	98.1%	✓ Verified
router	router	7	91.5%	✓ Verified
satellite dish device	satellite dish device	8	100.0%	✓ Verified
server rack	server rack	9	100.0%	✓ Verified

6. Software Engineering & Web Architecture

- **In-Memory Caching (@st.cache_resource):** Prevents the 22 MB model from reloading on every user interaction, dropping repeat inference time from 4.2s to under 80ms.
- **Decoupled Configuration:** Labels are isolated in class_names.json, enabling zero-code label taxonomy updates.
- **Two-Column Responsive UI:** Clean side-by-side presentation displaying the uploaded image alongside confidence metrics and top-5 probability breakdown progress bars.
- **Cloud Memory Optimization (tensorflow-cpu):** Replaced the heavy GPU build with the CPU-tailored distribution, avoiding out-of-memory crashes on free cloud hosting instances.

7. Deployment Guide & Production Status

Live Production Deployment: The application is deployed and publicly accessible on Streamlit Community Cloud at custom-image-classifier.streamlit.app.

Local Setup: Clone repository, install dependencies via `pip install -r requirements.txt`, and launch with `streamlit run app.py`.

Cloud Configuration: Environment operates on Python 3.11 with tensorflow-cpu for rapid server initialization and minimal memory consumption.

8. Future Scope & Conclusion

Future iterations can incorporate **YOLOv8** for multi-object detection and bounding boxes, live webcam stream classification via `st.camera_input`, and INT8 quantization using TensorFlow Lite for edge deployments on embedded processors.

In conclusion, the project successfully bridges deep learning theory and accessible web deployment, achieving a production-ready, high-accuracy classification pipeline for technology peripherals.